Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [18ASOE04]

# SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

## INTRODUCTION TO MICRO ELECTRO MECHANICAL SYSTEMS (MEMS)

**Time: 3 Hours** **Maximum Marks: 70**

**Instructions:**
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

| | | | |
|---|---|---|---|
| 1. | a. | Explain important applications of Micro Systems in microsatellites. | 07 |
| | b. | Explain Why Micro System requires multi-disciplinary approach ? | 07 |
| 2. | a. | Explain the application of Gas sensors. | 04 |
| | b. | Explain the wafer processing techniques.<br>i. Diffusion<br>ii. Ion Implantation | 10 |
| 3. | a. | Explain the silicon crystal growth process and Why silicon preferred compared to other materials. | 07 |
| | b. | Compare between the Wet etching and dry etching process used in micromachining. | 07 |
| | | OR | |
| 4. | a. | List out the special issues in micro system packaging. | 07 |
| | b. | Explain the physical deposition for depositing silicon dioxide. | 07 |
| 5. | a. | Explain the process involved in pulling CZ silicon bar. | 07 |
| | b. | Explain the advantages of using silicon for MEMS fabrication. | 07 |
| | | OR | |
| 6. | a. | Explain the different packaging materials used for MEMS device packaging. | 07 |
| | b. | Explain the advantages of using SU8 and PMMA for MEMS processing. | 07 |
| 7. | a. | Illustrate the major parameters tested in a Pressure sensor | 10 |
| | b. | List the application of MEMS Pressure sensor | 04 |
| | | OR | |
| 8. | a. | Explain any three reliability tests conducted on MEMS devices | 07 |
| | b. | Draw and explain the signal conditioning circuits used in pressure sensor integration. | 07 |